

IN-COURSE ASSESSMENT (ICA) SPECIFICATION

Module Title: Mobile App Development	Module Leader: Julien Cordry
	Module Code: CIS4034-N
Assignment Titles: Mobile App Development	Deadline Date: 6th January 2025
	Deadline Time: 4:00pm
	Submission Method: Online (Blackboard) ☒ Middlesbrough Tower ☐

Online Submission Notes:

- Please follow carefully the instructions given on the Assignment Specification
- When Extenuating Circumstances (e.g. extension) has been granted, a fully completed and signed Extenuating Circumstances form must be submitted to the School Reception or emailed to scedt-assessments@tees.ac.uk.

FULL DETAILS OF THE ASSIGNMENT ARE ATTACHED
INCLUDING MARKING & GRADING CRITERIA

Mobile App Development (CIS4034-N)

Assignment Specification

Introduction

You will design, plan and build a mobile app for a smartphone or tablet device using an industry-standard agile development method. Assessment will take the form of a presentation slide deck, supporting Android Emulator/Device screen recordings, with any additional documentation.

Students will submit their presentation materials, recordings and project artefacts (e.g. design, source code, app, documentation) as evidence.

The Android App

You will produce a single Android app using Kotlin and the Android SDK. ***The User Interface must be developed using Jetpack Compose wherever practically possible - where alternative UI technologies are used, which should be minimal, must be fully justified in the presentation.*** The app can be for any purpose, although you will need to discuss the size and scope of the app with your tutors. We don't expect an app store ready app, however it needs to demonstrate its intended behaviour. Unless otherwise agreed by the tutor, your app should aim to demonstrate the following:

Technical Expectations

1. Multiple screens

Your app should consist of multiple connected screens that respond suitably to the device screen size and orientation. You should include a splash screen as well. Data should be transmitted from one screen to another.

2. Mobile features

Your app should make good use of some mobile device features applicable to the needs of the app's purpose (e.g. location, camera, motion, fingerprint, etc.).

Suitable consideration should be given to the user experience, device performance and device battery life.

3. Data persistence

Your app should make use of one or more of the data persistence mechanisms available. If web/backend services (see next bullet) are used for primary data persistence, other local mechanisms should be used to cache data for the app's offline experience.

4. Use of web/backend services

Your app should use one or more web/backend services. These can be existing third-party services not created by yourself, including cloud-based data persistence services. If services are developed as part of this project, their code must be included in the submission.

5. Privacy, security, and permissions

Your app must not expose personal information about the user and should not be vulnerable to trivial attacks. Communications should be ciphered. Your application should not ask for more permissions than are needed.

The App

Marks for the app will be awarded for:

- The user experience,
- The mobile features,
- The potential interactions with a 3rd party service provider,
- The security and considerations given to the privacy of the users.

The Documentation

You must submit supporting some documentation. This should take the form of Agile documents detailing sprints undertaken with a sprint plan and a sprint review. The sprint review needs to have been checked by your tutor in class. You are expected to do five sprints during the course of the module.

The Presentation

You must prepare a 15-minute presentation about your app idea and working prototype – the resources for this will comprise of a small slide deck along with supporting product demonstration media (i.e.: screenshots and short recordings).

The presentation must be aimed at subject experts who have not previously been involved with your app development. You must demonstrate that you have considered the various professional, social, legal and ethical implications of the app, its development and its ultimate use.

A *substantial* portion of the presentation should be a demonstration of the working prototype. You will be expected to show the app running on an emulator or device through a screen recording.

Suggested Format

- Introduction of the App and target audience.
- App demonstration. Refer to screen recordings (see below), ideally embedded within the presentation or at least clearly labelled on submission.
- Justification of software design decisions (design and layout, hardware usage, user experience, data persistence, usage of 3rd party libraries etc.).
- Discuss any social, legal, ethical and security issues relevant to your app and its development and use.
 - For example, are you complying with the licensing requirements of the various Android and 3rd party libraries that you may be using?
 - Are you capturing and storing any personal/sensitive data? If so, are you complying with GDPR?
- The selection, use, and evaluation of an agile development method.
- The use of digital environments and tools for the management and execution of the project.
 - Project management tools, for example Trello.
 - Source code versioning management tools, for example Git.

Screen Recordings

Take multiple, short Android Emulator/Device screen recordings that each demonstrate a unique feature within your app. The idea here is to build evidence that your app behaves and functions as it should.

Deliverables

You must submit your work as a ZIP file via *Blackboard* with the following directory structure:

- **Source:** this directory should contain the entire source code for the project (i.e.: the Android app and any web/backend services), including any necessary project configuration files and third-party libraries.
- **Documentation:** this directory should contain the supporting document (*DOCX* or *PDF*), the presentation slide deck (*MS PowerPoint* format), Android Emulator/Device screen recordings, and any documentation of the product (e.g.: design diagrams).

Include a *readme* file in the root directory containing any necessary setup or configuration instructions.

Guidance

Embrace and follow the chosen agile development method - this is how professionals develop applications, and it is easier to follow it than to fake it.

Use a standard format for the package name of your application e.g.

uk.ac.tees.mad.studentnumber

Use source control (e.g.: Git) and other appropriate development/planning tools (e.g.: PivotalTracker, Trello, MS Excel). Use source control formally (i.e.: branch, merge, and tag) as appropriate to your development method.

Ensure that the documentation is in a suitable format for presentation and discussion at the presentations. Have diagrams and summary descriptors backed by more thorough explanatory text.

Learning Outcomes

Personal and Transferable Skills Development

- 1. Concisely present complex issues encountered during the development of a mobile app to an informed audience and defend the conclusions reached.*
- 2. Demonstrate sophisticated and professional use of digital environments and tools for the management, storage and documentation of software development projects.*
- 3. Make evidence-based decisions regarding software development methods and design issues, accepting accountability for the outcome of those decisions.*

Research, Knowledge and Cognitive Skills

- 4. Select and apply an industry-standard agile-oriented software development method, adapting it as necessary, during the development of a mobile app and critically evaluate its effectiveness.*

Professional Skills, Values and Behaviours

- 5. Apply industry-standard software development monitoring techniques to continuously adapt and improve own performance.*
- 6. Professionally manage the social, legal and ethical issues along with key risks associated with the development of a mobile app.*

Marking Criteria

Presentation	[10]	The app's purpose is explained clearly and its major features are stated. The target user is identified and described. The app is presented clearly and concisely and includes all the required information. Visual aids were used effectively during the presentation. Demonstration of the product is comprehensive.
Agile	[10]	2 points per sprint (validated by your module tutor). It is clear that an agile development method has been applied. There is plausible evidence of an appropriate number of successful development iterations (the commits match with the sprints).
Android app with source	[50]	The app builds successfully from source and can be deployed to an emulator or device. The app executes and functions correctly without causing uncaught exception. The technical expectations have been successfully demonstrated. The Android SDK has been used efficiently and effectively.
Tools usage	[10]	Demonstrable use of software development digital environments and tools in a professional and sophisticated manner covering version control and project planning. Evidence of tool evaluation and comparison also present.
Justification of software design decisions	[10]	Justification of software design decisions taken and defence of them in an academic and respectful manner covering design and layout, hardware usage, user experience, data persistence, and usage of 3rd party libraries.

Social, legal, ethical and security addressed	[10]	Appropriate consideration is given to the social, economic, legal, ethical and security issues associated with developing and using the proposed app.
---	------	---

Feedback

You should expect to receive your feedback within 20 working days of your deadline.

Mark ranges

You can expect the following marks for your submission.

Note: A simple, functional app is sufficient to achieve a pass.

Fail: 0–49%

- App: Missing, non-functional, or critically flawed. Does not meet basic requirements.
- Agile documentation: Lacks structure or relevance. No clear evidence of agile practices.
- Video: Missing or fails to demonstrate the app meaningfully.
- Presentation: Poorly delivered or lacks coherence.
- Overall: Does not meet the minimum academic or technical standards.

Pass: 50–59%

- App: A simple but functional app that meets core requirements. May have minor bugs or limited features.
- Agile documentation: Basic agile documentation showing some planning and iteration.
- Video: Demonstrates the app clearly, even if briefly.

- Presentation: Communicates the development process and outcomes adequately.
- Overall: Meets minimum expectations. Demonstrates understanding and effort, even if limited in scope or polish.

Merit: 60–69%

- App: Well-structured and functional with thoughtful design. Good use of Kotlin and Jetpack Compose.
- Agile documentation: Clear and consistent agile documentation with evidence of iterative development and reflection.
- Video: Well-produced and informative, not too much insistence on basic things such as log in.
- Presentation: Confident and well-organised, showing good understanding of design and development choices.
- Overall: Strong submission with good technical and academic standards.

Distinction: 70–100%

- App: Highly polished, feature-rich, and possibly innovative. Excellent use of Kotlin and Jetpack Compose.
- Agile documentation: Comprehensive, insightful, and well-aligned with agile principles, matching the work done with git.
- Video: Professional quality, clearly showcasing features and user experience.
- Presentation: Engaging, articulate, and demonstrates deep understanding and critical thinking.
- Overall: Outstanding work that exceeds expectations in all areas.